

Not all hyperbolic 4-manifolds are pseudo-arithmetic

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Abstract

We exhibit three finite-volume hyperbolic Coxeter lattices in dimension four whose adjoint trace fields are not totally real. They are the reflection groups of the Ma–Zheng polytopes $P_{11,6}, P_{11,7}, P_{11,8}$. The obstruction is visible in only two cyclic products of $A = 2G$, where G is the unit-normal Gram matrix. These products generate elements $s = \sqrt{5}$ and u satisfying

$$(s - 1)u^2 + 2(s - 3)u + s - 7 = 0.$$

At the conjugate place $s \mapsto -s$, the discriminant is $8(1 - \sqrt{5}) < 0$. Thus the adjoint trace field has a complex place, whereas the adjoint trace field of a pseudo-arithmetic lattice in dimension greater than three is totally real. Passing to torsion-free orientation-preserving finite-index subgroups gives noncompact finite-volume hyperbolic 4-manifolds that are not pseudo-arithmetic. This answers two questions of Emery–Mila in the negative.

1 Introduction

The adjoint trace field and the ambient algebraic group are basic commensurability invariants of a finite-volume hyperbolic manifold. Emery and Mila introduced pseudo-arithmetic lattices to encompass both arithmetic lattices and the nonarithmetic lattices produced by the standard gluing constructions [2]. They asked whether every lattice in $\mathrm{PO}(n, 1)$, for $n > 3$, is pseudo-arithmetic and, more weakly, whether its adjoint trace field must be totally real [2, Questions 1.8 and 1.10].

Both questions have negative answers already in dimension four. The examples come from Ma and Zheng’s classification of the 331 finite-volume hyperbolic Coxeter 4-polytopes with seven facets [5]. We use three consecutive rows of their table.

Theorem 1. *Let $P_{11,j}$, $j = 6, 7, 8$, be the Ma–Zheng finite-volume hyperbolic Coxeter 4-polytopes described in Table 1, and let $\Gamma_{11,j} < \mathrm{PO}(4, 1)$ be their reflection groups. Then the adjoint trace field of every $\Gamma_{11,j}$ is not totally real. Consequently none of these lattices is pseudo-arithmetic.*

Corollary 1. *There exist orientable, noncompact, finite-volume hyperbolic 4-manifolds that are not pseudo-arithmetic.*

The proof of Theorem 1 is deliberately short. For the normalized Gram matrix G , put $A = 2G$. One two-cycle in A recovers $\sqrt{5}$, and one three-cycle recovers an element u . A quadratic relation for u then has negative discriminant at the nontrivial real place of $\mathbb{Q}(\sqrt{5})$. This argument uses the field generated by cyclic products, which is the adjoint trace field of a finite-volume Coxeter reflection group; it does not confuse that field with the generally larger field generated by the individual Gram entries.

Emery–Mila verified pseudo-arithmeticity for the higher-dimensional Coxeter groups available to them, and work published in 2026 still described all then-known lattices in $\mathrm{PO}(n, 1)$, $n > 3$, as

pseudo-arithmetic [1]. We know of no previously recorded non-pseudo-arithmetic lattice in these dimensions. This records a targeted literature search, rather than claiming that any bounded search proves absolute priority.

2 Trace fields and pseudo-arithmeticity

Let $n > 3$, and let $\Gamma < \mathrm{PO}(n, 1)$ be a lattice. Its *adjoint trace field* is

$$k(\Gamma) = \mathbb{Q}(\{\mathrm{tr}(\mathrm{Ad} \gamma) : \gamma \in \Gamma\}) \subset \mathbb{R},$$

where Ad is the adjoint representation. This is a number field. Vinberg proved that it is a commensurability invariant and that there is, up to $k(\Gamma)$ -isomorphism, a unique $k(\Gamma)$ -form $\mathbf{G}_\Gamma$ of the relevant real projective orthogonal group for which $\Gamma \subset \mathbf{G}_\Gamma(k(\Gamma))$ [7]. The group $\mathbf{G}_\Gamma$ is called the *ambient group* of Γ .

For a number field $k \subset \mathbb{R}$, a k -form $\mathbf{H}$ of $\mathrm{PO}(n, 1)$ is *admissible* if $\mathbf{H}(k \otimes_{\mathbb{Q}} \mathbb{R})^\circ$ has exactly one noncompact factor, and that factor is isomorphic to $\mathrm{PO}(n, 1)^\circ$. When $n > 3$, admissibility forces k to be totally real. Following Emery–Mila, a lattice with adjoint trace field K is *pseudo-arithmetic* if there are an admissible k -group $\mathbf{H}$ and a finite real multiquadratic extension

$$K = k(\sqrt{d_1}, \dots, \sqrt{d_t}), \quad d_i \in k^\times,$$

such that $\mathbf{G}_\Gamma \cong_K \mathbf{H}_K$. Here *real* is part of the pseudo-admissibility condition: all archimedean embeddings of K are real. The possibly trivial case $t = 0$ includes quasi-arithmetic lattices.

We will need only the following immediate obstruction.

Proposition 1 (Emery–Mila). *If $n > 3$ and $\Gamma < \mathrm{PO}(n, 1)$ is pseudo-arithmetic, then its adjoint trace field $k(\Gamma)$ is totally real.*

Proof. In the notation of the definition, the field k is totally real by admissibility and K/k is a real multiquadratic extension. Hence every embedding $K \hookrightarrow \mathbb{C}$ has image in $\mathbb{R}$. This consequence is also stated explicitly in [2, Section 1.4]. $\square$

We next recall the reflection-group description of the same field. Let $P \subset \mathbb{H}^n$ be a finite-volume Coxeter polytope with facets $F_0, \dots, F_{N-1}$. Choose outward unit spacelike normals e_i , and write

$$G = (g_{ij}), \quad g_{ij} = \langle e_i, e_j \rangle.$$

Thus $g_{ii} = 1$; if F_i, F_j meet at angle π/m_{ij} , then $g_{ij} = -\cos(\pi/m_{ij})$; parallel facets have $g_{ij} = -1$; and ultraparallel facets at distance ρ_{ij} have $g_{ij} = -\cosh \rho_{ij}$. Put $A = 2G = (a_{ij})$. A *cyclic product* is an expression

$$a_{i_0 i_1} a_{i_1 i_2} \cdots a_{i_{m-1} i_m} a_{i_m i_0}.$$

The *Vinberg field* $k(P)$ is the field generated over $\mathbb{Q}$ by all such products.

Proposition 2 (Vinberg). *If P has finite volume and $\Gamma(P)$ is its reflection group, then*

$$k(\Gamma(P)) = k(P).$$

This is the cyclic-product form of Vinberg’s field-of-definition theorem [7]; see also [2, Section 4.1] and the modern formulation in [4, Section 3.1]. The normalization matters only for bookkeeping: Emery–Mila use the bilinear Gram matrix with diagonal 2, which is precisely $A = 2G$ in the unit-normal convention above.

3 The three Coxeter polytopes

We use zero-based facet indices. A Coxeter vector on seven facets records the 21 pairs in lexicographic blocks:

$$(01, 02, 03, 04, 05, 06) \mid (12, 13, 14, 15, 16) \mid (23, 24, 25, 26) \mid (34, 35, 36) \mid (45, 46) \mid (56). \quad (1)$$

An integer $m \geq 2$ records a dihedral angle π/m ; the symbol 0 records a parallel pair; and the letters a, b record ultraparallel pairs with $g_{ij} = -a, -b$, respectively. Thus, for example, the initial block $2232ab$ says

$$m_{01} = 2, \quad m_{02} = 2, \quad m_{03} = 3, \quad m_{04} = 2, \quad g_{05} = -a, \quad g_{06} = -b.$$

The three rows of the Ma–Zheng census are as follows. The volume column is the volume of the Coxeter fundamental polytope.

polytope	Coxeter vector	cusps	volume
$P_{11,6}$	2232ab 52322 2220 442 22 2	1	$13\pi^2/864$
$P_{11,7}$	2232ab 52323 2220 442 22 2	1	$\pi^2/54$
$P_{11,8}$	2232ab 52324 2220 442 22 2	2	$19\pi^2/864$

Table 1: Three finite-volume Coxeter 4-polytopes from Ma–Zheng [5].

Put

$$s = \sqrt{5}, \quad r = \sqrt{2}.$$

The three rows have the same first ultraparallel entry

$$a = \frac{1}{2}\sqrt{7 + s + 2\sqrt{2 + 2s}}. \quad (2)$$

For completeness, their other ultraparallel entries are

$$\begin{aligned} b_6 &= \sqrt{\frac{9}{2} + 2s + \sqrt{38 + 17s}}, \\ b_7 &= r + \frac{\sqrt{10}}{2} + \frac{1}{2}\sqrt{29 + 13s}, \\ b_8 &= \frac{1}{4}(2 + 3r + 2s + rs + 2\sqrt{19 + 14r + 9s + 6rs}). \end{aligned} \quad (3)$$

The positive radicals in (2)–(3) are the values tabulated by Ma–Zheng; in particular $a, b_j > 1$, as required for ultraparallel facets. Only the common entry a will enter the obstruction.

Reading (1) from any of the three vectors gives

$$A_{12} = -2 \cos \frac{\pi}{5} = -\frac{1+s}{2}, \quad A_{03} = -1, \quad A_{35} = -r, \quad A_{50} = -2a. \quad (4)$$

Equations (2) and (4) contain all the Coxeter data used in the proof.

4 The two-cyclic-product obstruction

We isolate the elementary number-field calculation first.

Lemma 1. *Let $s = \sqrt{5}$, let a be given by (2), and put $u = \sqrt{2}a$. Then*

$$(s-1)u^2 + 2(s-3)u + s - 7 = 0. \quad (5)$$

Moreover, the number field $L = \mathbb{Q}(s, u)$ is not totally real.

Proof. Set $t = \sqrt{2+2s}$. From (2),

$$u^2 = 2a^2 = \frac{7+s}{2} + t.$$

Using $s^2 = 5$, a direct calculation gives

$$\left(\frac{3-s+t}{s-1}\right)^2 = \frac{7+s}{2} + t = u^2.$$

Since $1 < s < 3$ and $t > 0$, both u and $(3-s+t)/(s-1)$ are positive. Hence

$$(s-1)u = 3-s+t.$$

Squaring $t = (s-1)u + s - 3$, using $t^2 = 2 + 2s$, and dividing by $s-1 \neq 0$ gives (5).

Consider (5) as a quadratic polynomial $f(X) \in \mathbb{Q}(s)[X]$. Its discriminant is

$$\begin{aligned} \Delta &= 4(s-3)^2 - 4(s-1)(s-7) \\ &= 8(1+s). \end{aligned} \quad (6)$$

Let $\sigma : \mathbb{Q}(s) \hookrightarrow \mathbb{R}$ be the nontrivial embedding, $\sigma(s) = -s$. The conjugate polynomial f^σ has discriminant

$$\sigma(\Delta) = 8(1-s) < 0. \quad (7)$$

In particular f^σ has no real root.

If u belonged to $\mathbb{Q}(s)$, then the real number $\sigma(u)$ would be a root of f^σ , contradicting (7). Thus f is the minimal polynomial of u over $\mathbb{Q}(s)$. Extending σ to L sends u to one of the two nonreal roots of f^σ . Hence L has a complex place and is not totally real. $\square$

Proof of Theorem 1. Fix $j \in \{6, 7, 8\}$, and put

$$K_j = k(\Gamma_{11,j}) = k(P_{11,j}),$$

where the equality is Proposition 2. From (4), two cyclic products in $A = 2G$ are

$$\begin{aligned} c_{12} &= A_{12}A_{21} = \frac{3+s}{2}, \\ c_{035} &= A_{03}A_{35}A_{50} = -2ra. \end{aligned} \quad (8)$$

Consequently

$$s = 2c_{12} - 3 \in K_j, \quad u = ra = -\frac{1}{2}c_{035} \in K_j.$$

Thus $L = \mathbb{Q}(s, u)$ is a subfield of K_j . By Lemma 1, L has a nonreal embedding $\iota : L \hookrightarrow \mathbb{C}$. Since K_j/L is a finite separable extension, ι extends to an embedding $K_j \hookrightarrow \mathbb{C}$. Its image cannot lie in $\mathbb{R}$, because its restriction to L does not. Hence K_j is not totally real.

Proposition 1 now shows that $\Gamma_{11,j}$ is not pseudo-arithmetic. The same argument applies to all three values of j . $\square$

Proof of Corollary 1. Let $\Gamma = \Gamma_{11,j}$. Its orientation-preserving subgroup $\Gamma^+ = \Gamma \cap \text{Isom}^+(\mathbb{H}^4)$ has index two. Since a Coxeter group is a finitely generated linear group, Selberg's lemma [6] gives a torsion-free finite-index subgroup $\Lambda < \Gamma^+$. Therefore

$$M_j = \Lambda \backslash \mathbb{H}^4$$

is an orientable finite-volume hyperbolic 4-manifold. It is noncompact because $P_{11,j}$ has at least one ideal vertex. Adjoint trace fields are invariant under passage to finite-index subgroups [7]; hence $k(\Lambda) = k(\Gamma) = K_j$. Since K_j is not totally real, Proposition 1 shows that Λ , and thus M_j , is not pseudo-arithmetic. $\square$

5 Exact certification and reproducibility

The proof above is hand-checkable and requires neither a primitive element for the full trace field nor a numerical signature calculation. We nevertheless record independent exact checks, both to validate the transcription of the census rows and to make the examples reproducible.

5.1 Gram matrices

For each of the three rows, exact elimination verifies that the seven-by-seven unit-normal Gram matrix has rank 5. All seven principal six-by-six minors vanish. The principal block on facets 0, 1, 2, 3, 4 has leading principal minors

$$1, \quad 1, \quad \frac{5-s}{8}, \quad \frac{3(5-s)}{32}, \quad -\frac{1+s}{32}. \quad (9)$$

Thus this block has inertia $(4, 1)$, and the full Gram matrix has exact inertia

$$(n_+, n_-, n_0) = (4, 1, 2).$$

The calculation also verifies (2)–(5) symbolically. Ma–Zheng's classification supplies finite volume for all three rows.

For $P_{11,8}$, we additionally checked finite volume directly from the principal subdiagrams. Among the ten maximal elliptic or parabolic subdiagrams, eight are elliptic of rank four. The other two are the rank-three parabolic diagrams on

$$\{1, 3, 4, 6\} \quad \text{and} \quad \{2, 3, 4, 5, 6\};$$

their types are, respectively, $\tilde{C}_3$ and $\tilde{A}_1 \sqcup \tilde{C}_2$. The latter has nullity two and is the Euclidean triangular-prism link of the nonsimple ideal vertex. Every one of the 21 elliptic rank-three edge diagrams has exactly two ordinary or ideal endpoints. Vinberg's finite-volume criterion [8] therefore gives two cusps and finite covolume. Independently, the pinned CoxIter calculation [3] returns

$$(f_0, f_1, f_2, f_3, f_4) = (10, 21, 18, 7, 1), \quad \text{vol}(P_{11,8}) = \frac{19\pi^2}{864}.$$

5.2 A full-field check

The two cycles in (8) are the smallest certificate, but the whole adjoint trace field of $P_{11,8}$ can also be recovered. In addition to them, the cycles

$$c_{126} = -r(1+s), \quad c_{03416} = -4b_8$$

successively recover r, a, b_8 . Since every Gram entry belongs to $\mathbb{Q}(s, r, a, b_8)$, this proves

$$k(P_{11,8}) = \mathbb{Q}(s, r, a, b_8).$$

Set $w = 2(r + a + b_8)$. Exact power-basis expressions for s, r, a, b_8 in terms of w show that w is primitive, so $k(P_{11,8}) = \mathbb{Q}(w)$. Its minimal polynomial is

$$\begin{aligned} p_w(x) = & x^8 - 8x^7 - 252x^6 - 1392x^5 - 3524x^4 \\ & - 5280x^3 - 5712x^2 - 3968x - 1136. \end{aligned} \tag{10}$$

An independent PARI/GP computation gives number-field signature $(r_1, r_2) = (4, 2)$ and discriminant

$$40960000 = 2^{16}5^4.$$

In particular, the complete field computation agrees with the much shorter complex-place certificate in Lemma 1.

5.3 Artifacts and verification boundary

The accompanying repository¹ contains the following exact verification artifacts:

- `scripts/p11_6_7_non_total_trace.py` and `scripts/p11_8_non_total_trace.py` reproduce the exact rank, inertia, cyclic-product, and field calculations; the latter invokes PARI/GP to check irreducibility, signature, and discriminant for (10);
- `scripts/p11_8_finite_volume.py` exhausts the 127 nonempty principal submatrices and checks the vertex and edge conditions;
- `scripts/p118_trace_field.py` reconstructs the Vinberg form by a second primitive-element route, while `scripts/p118_field_independent.gp` checks irreducibility, the real-root count, the field signature, and the discriminant of the resulting alternative degree-eight polynomial;
- the corresponding machine-readable certificates are stored in `results/`, together with a pinned CoxIter input, captured output, and build provenance.

The Ma–Zheng TeX source used for the three rows has SHA-256

$$0017931637f017641d4935450cc2a07d070866d9c749443d9e3ede9f3a39f003.$$

Each symbolic case is run in a fresh process; the observed peak resident set size is below 100 MB.

The directory `formal/` is a pinned Lean 4/mathlib project. It kernel-checks the intrinsic algebraic statement that, for a number field K and $s, u \in K$,

$$\text{minpoly}_{\mathbb{Q}}(s) = X^2 - 5 \quad \text{and} \quad (s - 1)u^2 + 2(s - 3)u + s - 7 = 0 \quad \implies \quad K \text{ is not totally real.}$$

In particular, Lean obtains the embedding $s \mapsto -\sqrt{5}$ from the minimal polynomial and proves that it cannot be real. A second module kernel-checks the normalized Clifford and split-support implication chains conditionally: ABHN, Hasse–Minkowski, local Brauer functoriality, and the pseudo-arithmetic characterization occur as explicit theorem hypotheses, not as new axioms. The Ma–Zheng classification and radical data, the concrete extraction of s, u from the Coxeter cycles, Vinberg’s cyclic-product theorem, and the Emery–Mila total-reality implication likewise remain named inputs at the application boundary. The development contains no project-defined axiom or opaque declaration and no `sorry` or `admit`.

¹<https://github.com/sashakolpakov/PseudoArithmetic>

References

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